Paper IV – Recent Advances (MCh Urology, Winter 2017)

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IV-1. Discuss Robot and its role in urology.

Type: LAQ (25 marks)

Introduction

- Surgical robots are **master–slave telemanipulators**: the surgeon controls every movement; the robot acts on nothing by itself.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, so RARP (robot-assisted radical prostatectomy) spread where laparoscopic prostatectomy had stalled.

History

- **PUMA 560 (1985)** first surgical robot; **PROBOT** first urological robot (TURP (transurethral resection of prostate)).
- **da Vinci** FDA (Food and Drug Administration) approval and first RARP in **2000**; first paediatric robotic procedure **2002**.

da Vinci system

1. **Surgeon console**: 3D (three-dimensional) magnified view, motion scaling, tremor filtration.
2. **Patient-side cart**: **EndoWrist** instruments with **7 DOF (degrees of freedom)**; ports 8 mm, 8–10 cm apart to avoid arm collisions.
3. **Vision cart**: light source, camera electronics, energy generator.

Generation	Added feature
S (2006) / Si	High definition, fourth arm, dual console for training
Xi (2014)	8-mm camera in any port, finer arms, docks from either side, no stirrups
SP (2018)	All instruments through one 2.5-cm cannula

- After patent expiry, other platforms appeared (e.g. **Hugo, Versius**, the Indian **SSI Mantra**).

Advantages and disadvantages

Advantages	Disadvantages
3D vision, wristed instruments, easy suturing	High capital and instrument cost
Tremor filtration, seated ergonomics	Limited haptic feedback
Less blood loss and shorter stay than open; fewer conversions than laparoscopy	Docking time, bulk, surgeon away from patient

Applications (with evidence)

1. RARP

- **Yaxley (Lancet 2016)**, RCT (randomised controlled trial): continence and potency **equal to open at 12 weeks**, less blood loss, shorter stay; equal at **24 months** on follow-up.
- Positive margins equal; the surgeon, not the platform, is the main variable.
- **Retzius-sparing** and **hood** techniques preserve continence structures and give earlier continence.

2. RAPN (robot-assisted partial nephrectomy)

- Versus laparoscopy: shorter warm ischaemia and fewer conversions, so more complex and hilar tumours are done minimally invasively.
- **Trifecta** = negative margin, no major complication within 30 days, warm ischaemia ≤25 minutes.

3. RARC (robot-assisted radical cystectomy)

- **RAZOR (Lancet 2018)**: **non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), less blood loss and transfusion.
- **iROC**: robotic cystectomy with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**, fewer VTE (venous thromboembolism) and wound complications.

4. Other procedures

Procedure	Why the robot helps
Pyeloplasty, ureteric reimplantation	Easy intracorporeal suturing; also in children (small working space)
BMG (buccal mucosa graft) ureteroplasty	Long proximal or mid-ureteric strictures
RPLND (retroperitoneal lymph node dissection)	Template, nerve-sparing; short stay
Adrenalectomy, nephroureterectomy	Equivalent to laparoscopy
Simple prostatectomy	Large glands; less bleeding than open
Sacrocolpopexy, VVF (vesicovaginal fistula) repair	Deep pelvic suturing
RAKT (robot-assisted kidney transplantation)	Fewer wound complications in obese recipients

Paediatric robotics

- Reconstruction (pyeloplasty, reimplantation, appendicovesicostomy) is where paediatric robotics has grown. Lower insufflation pressure; "burp" ports to enlarge the working space.

Physiology and positioning

- **CO_2 pneumoperitoneum** (12–15 mm Hg, lower in children) causes hypercarbia, acidosis, reduced venous return and oliguria.
- **Steep Trendelenburg** for pelvic surgery raises intracranial and intraocular pressure.

Complications specific to robotics

Problem	Details
Steep Trendelenburg	Facial and laryngeal oedema, raised IOP (intraocular pressure), corneal abrasion
Nerve injury	Brachial plexus (avoid shoulder braces); femoral, peroneal
Compartment syndrome, rhabdomyolysis	Long lithotomy or flank position
Thermal bowel injury	Insulation failure, capacitive coupling; presents late
Robot malfunction	<1%; half found before the patient enters
Port-site hernia	Close larger port sites

- **Prevention**: pad pressure points, secure arms before tilting, limit tilt and lithotomy time, protect the eyes.

Training and credentialing

- Box trainer, **VR (virtual reality)** simulators, animal and cadaver labs, then proctored cases. **FRS (Fundamentals of Robotic Surgery)** curriculum. Outcomes improve with volume.

Recent advances

- Single-port transvesical and extraperitoneal surgery; force feedback (da Vinci 5); **AI (artificial intelligence)** video-based skill assessment; augmented reality 3D overlay; **telesurgery** (limited by latency).
- **Also know:** ERAS (enhanced recovery after surgery) with same-day RARP discharge; ICG (indocyanine green) fluorescence for selective clamping and ureteric perfusion.

Diagram to draw: console, vision cart and patient cart; RARP port placement.

Key points to remember:

- da Vinci 2000; Xi 2014; SP 2018; 7 degrees of freedom.
- Yaxley: RARP equal to open at 12 weeks; RAZOR non-inferior; iROC more days out of hospital.
- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Trendelenburg causes oedema and raised IOP; avoid shoulder braces.

IV-2. Discuss pathophysiology of allograft rejection. Discuss newer immunosuppressive drugs in renal transplant.

Type: LAQ (25 marks)

Introduction

- Rejection is the recipient's immune response against donor (allo) antigens. Acute rejection has fallen from over 60% to **below 10%**; slow, subclinical **antibody injury** is now the main problem.

Part A: Pathophysiology of rejection

Target antigens

- **HLA (human leucocyte antigen) / MHC (major histocompatibility complex)**: class I (A, B, C) on all nucleated cells; class II (DR, DQ, DP) on antigen-presenting and endothelial cells.
- **ABO** antigens; minor and non-HLA antigens. **Sensitisation** (transfusion, pregnancy, previous transplant) creates preformed anti-HLA antibodies.

Allorecognition (how T cells see the graft)

Pathway	Mechanism	Main role
Direct	Recipient T cells recognise **intact donor MHC** on donor "passenger" dendritic cells	Strong early acute TCMR (T-cell-mediated rejection)
Indirect	Recipient APCs (antigen-presenting cells) present processed **donor peptides on self-MHC**	Lifelong; helps B cells, so alloantibody and chronic rejection
Semi-direct	Recipient APCs acquire **intact donor MHC** (membrane transfer) and present it	Links the two pathways

- **Ischaemia–reperfusion injury** activates innate immunity and up-regulates MHC, so DGF (delayed graft function) raises rejection risk.

Three-signal model of T-cell activation

1. **Signal 1**: TCR (T-cell receptor) binds MHC–peptide → **calcineurin** → IL-2 (interleukin-2) transcription.
2. **Signal 2 (costimulation)**: **CD28 binds CD80/86** (B7). Signal 1 without signal 2 causes anergy.
3. **Signal 3**: IL-2 binds **CD25** (IL-2 receptor) → **mTOR (mammalian target of rapamycin)** → G1 to S → proliferation (needs purine synthesis).

Effector mechanisms

- **Cellular**: CD8 cytotoxic T cells (perforin, granzyme) and CD4 Th1 cells with macrophages cause **tubulitis** and **endarteritis**.
- **Humoral**: **DSA (donor-specific antibody)** binds endothelium → complement, leaving **C4d** on peritubular capillaries → **microvascular inflammation** (glomerulitis, capillaritis). NK (natural killer) cell injury is complement-independent, so AMR (antibody-mediated rejection) may be C4d-negative.

Types of rejection

Type	Timing	Mechanism and features
Hyperacute	Minutes–hours	Preformed ABO/HLA antibody → complement, thrombosis; graft turns blue; **prevented by crossmatch**

Type	Timing	Mechanism and features
Accelerated acute	2–5 days	Memory (anamnestic) response in sensitised recipients
Acute TCMR	Days–months (mostly first 6 months)	Interstitial infiltrate, tubulitis, arteritis
Acute AMR	Any time	DSA, C4d, microvascular inflammation
Chronic active AMR	Months–years	**Transplant glomerulopathy** (GBM (glomerular basement membrane) double contours), arterial intimal fibrosis; commonest cause of late graft loss

- **Also know:** subclinical rejection (histology, stable function). **Risk factors**: HLA mismatch, DSA, ABO incompatibility, long ischaemia and DGF, **non-adherence**, retransplant.

Banff classification (biopsy is the gold standard)

Category	Key criteria
Borderline	Foci of tubulitis with mild interstitial inflammation
TCMR IA / IB	Interstitial inflammation >25% with moderate (IA) or severe (IB) tubulitis
TCMR IIA / IIB	Mild-moderate (IIA) or severe (IIB) intimal arteritis
TCMR III	Transmural arteritis or fibrinoid necrosis
Active AMR	All three: (1) tissue injury (microvascular inflammation, arteritis, TMA (thrombotic microangiopathy)); (2) antibody–endothelium interaction (C4d, microvascular inflammation or gene transcripts); (3) **circulating DSA**

Diagnosis and monitoring

- Rising creatinine; **de novo DSA** predicts graft loss; **dd-cfDNA (donor-derived cell-free DNA)** (negative predictive value >90%) avoids unnecessary biopsy; gene-expression panels.
- **Exclude**: obstruction, leak, renal artery stenosis, CNI (calcineurin inhibitor) toxicity, BK virus nephropathy.

Treatment

- **TCMR**: IV (intravenous) pulse steroids; **rATG (rabbit antithymocyte globulin)** for Banff II–III.
- **AMR** (not standardised): steroids, **IVIG (intravenous immunoglobulin)**, **plasmapheresis**, rituximab; eculizumab in selected patients.

Part B: Newer immunosuppressive drugs

- Standard: induction plus **tacrolimus + mycophenolate** ± steroids. Newer drugs aim to avoid CNI nephrotoxicity, improve adherence and treat AMR.

Drug	Mechanism	Key points
Belatacept	**CTLA4-Ig**: binds CD80/86, blocks **signal 2**	IV monthly; **BENEFIT**: better GFR (glomerular filtration rate), more early rejection; **contraindicated if EBV (Epstein–Barr virus)-seronegative** (PTLD (post-transplant lymphoproliferative disorder))
mTOR inhibitors (sirolimus, everolimus)	Block mTOR (**signal 3**)	Not nephrotoxic; fewer CMV (cytomegalovirus) infections; **TRANSFORM**: everolimus + low-dose CNI non-inferior; poor wound healing, **lymphocele**

Drug	Mechanism	Key points
Extended-release tacrolimus	CNI (**signal 1**)	Once daily, better adherence
Basiliximab	Anti-CD25 (IL-2 receptor), non-depleting	Low-risk induction, days 0 and 4
rATG; alemtuzumab	T-cell depletion; anti-CD52	High-risk induction; infection, PTLD
Rituximab; bortezomib	Anti-CD20 (B cells); proteasome inhibitor (plasma cells)	AMR, desensitisation
Eculizumab	Anti-C5	Selected AMR; aHUS (atypical haemolytic uraemic syndrome)
Imlifidase	Enzyme that **cleaves IgG**	EMA (European Medicines Agency) 2020: crossmatch-positive, highly sensitised
Anti-IL-6 (tocilizumab, clazakizumab)	Blocks IL-6	Chronic active AMR

Diagram to draw: three-signal T-cell activation with drug sites (CNI at signal 1, belatacept at 2, basiliximab and mTOR inhibitors at 3, mycophenolate at DNA synthesis).

Key points to remember:

- Direct pathway drives early TCMR; indirect pathway drives antibody and chronic rejection.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept: better GFR, avoid if EBV-negative; imlifidase for highly sensitised.

IV-3a. Botulinum toxin in urology

Type: SAQ (10 marks)

See Paper III, Q3b (full answer).

Key points to remember (recent advances angle):

- Paediatric NDO (neurogenic detrusor overactivity): FDA approved onabotulinumtoxinA for children aged ≥5 years (2021); used from about 2 years of age.
- Other formulations: abobotulinumtoxinA and incobotulinumtoxinA under study for detrusor overactivity; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.
- Standard adult intradetrusor doses for comparison: onabotulinumtoxinA 100 U (idiopathic OAB (overactive bladder)), 200 U (NDO).

IV-3b. Sacral Neuromodulation

Type: SAQ (10 marks)

See Paper III, Q3a (full answer).

Key points to remember (recent advances angle):

- Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
- Full-body MRI (magnetic resonance imaging)-conditional devices at 1.5 T and 3 T.
- Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
- Standardised tined-lead placement improves outcomes and lowers revision.

IV-3c. GNRH Analogue

Type: SAQ (10 marks)

Read as: GnRH (gonadotrophin-releasing hormone, also LHRH, luteinising hormone-releasing hormone) agonists and antagonists, chiefly in prostate cancer.

Physiology

- **Pulsatile** GnRH stimulates pituitary LH (luteinising hormone) and FSH (follicle-stimulating hormone); LH drives Leydig-cell testosterone (about 95% of total). **Continuous** stimulation desensitises the receptor: the basis of agonist therapy.

GnRH agonists

- **Drugs**: leuprolide, goserelin, triptorelin, histrelin. Position-6 substitution raises potency; depots last 1 month to 1 year.
- **Mechanism**: initial LH surge, then receptor **down-regulation** → low LH → castration in 2–4 weeks; survival equals orchidectomy.
- **Flare**: LH up to 10-fold for 10–20 days. Dangerous with spinal or weight-bearing metastases (cord compression, pain). **Prevent with an antiandrogen** (e.g. bicalutamide) started with or before the agonist, for **21–28 days**.

GnRH antagonists

- Immediate competitive receptor block: LH falls 84% in 24 hours, **no flare**, FSH falls.
- **Degarelix** (SC (subcutaneous) monthly): castrate by day 3 in >95%; injection-site reactions.
- **Relugolix** (oral daily, FDA 2020). **HERO** vs leuprolide: sustained castration 97% vs 89%; faster recovery after stopping; **fewer MACE (major adverse cardiovascular events)** (about 3% vs 6%).
- Preferred for **impending cord compression** or severe bone pain when orchidectomy is unsuitable.
- **Also know:** abarelix restricted (severe allergy); PRONOUNCE (degarelix vs leuprolide, stopped early) showed no MACE difference.

Castrate level

- **<50 ng/dL** (traditional); **<20 ng/dL** is achievable (orchidectomy median about 15) and may be better. Check testosterone when PSA (prostate-specific antigen) rises.

Indications in urology

- Metastatic, locally advanced (with radiotherapy) and recurrent prostate cancer; continued in CRPC (castration-resistant prostate cancer).
- **Also know:** BPH (benign prostatic hyperplasia) volume reduction (not routine); central precocious puberty; cryptorchidism (GnRH or hCG (human chorionic gonadotrophin): low success, not first-line); endometriosis.

Adverse effects

- Hot flushes (50–80%), loss of libido, ED (erectile dysfunction), **osteoporosis** (denosumab or zoledronic acid), diabetes, cardiovascular events, anaemia, sarcopenia.

Diagram to draw: hypothalamic–pituitary–testis axis showing agonist and antagonist sites.

Key points to remember:

- Agonist: surge, then down-regulation; cover flare with antiandrogen 21–28 days.
- Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE).
- Castrate <50 ng/dL; aim <20 ng/dL.

IV-3d. Miniperc

Type: SAQ (10 marks)

Definition

- PCNL (percutaneous nephrolithotomy) through a **smaller tract** than standard, aiming to reduce bleeding and renal trauma. First "mini-perc" by **Jackman (1998)**, 13 Fr, in children.

Designation	Tract size
Standard	24–34 Fr
Mini	15–22 Fr
Ultra-mini	11–14 Fr
Micro-perc	<10 Fr (single-step "all-seeing needle")

- The nomenclature is contested. The **working channel**, not sheath size, decides which instruments pass.

Instruments and technique

- Mini nephroscopes (e.g. 12 Fr telescope, 15–18 Fr sheath, 6 Fr channel); laser (holmium/thulium) or small pneumatic probes.
- **Fragment removal**: irrigation outflow around the scope washes fragments out (**"vacuum-cleaner" effect**), or **active suction sheaths** (super-mini PCNL).
- Ultrasound or fluoroscopic access; supine or prone; ECIRS (endoscopic combined intrarenal surgery) possible.
- Often **tubeless** or small nephrostomy (≤10 Fr: less pain).

Advantages over standard PCNL

- Less haemoglobin drop, **fewer transfusions**, shorter stay; less parenchymal damage; useful in children.

Disadvantages

- **Longer operative time**, poorer visualisation, slower fragment clearance.
- **Intrarenal pressure**: high-pressure irrigation through a small sheath, especially when the scope nearly fills it, can raise pelvic pressure → pyelovenous backflow, **fever and sepsis**. Keep outflow free, use low-pressure irrigation, or suction sheaths.

Evidence

- Mini vs standard (meta-analysis): **non-inferior stone-free rate**, less bleeding and stay, longer surgery; same complications.
- Stones >2 cm: equal stone-free rate and fever; mini bleeds less than **30 Fr** tracts, but **every advantage disappears versus 24 Fr**.

Indications

- Renal stones **1–2 cm**, **lower pole stones** (alternative to RIRS (retrograde intrarenal surgery)), failed SWL (shock wave lithotripsy), **paediatric stones**, calyceal diverticular stones; selected larger stones.
- **Contraindications**: as for PCNL (untreated infection, coagulopathy, pregnancy).

Diagram to draw: tract sizes compared (standard, mini, ultra-mini, micro).

Key points to remember:

- Mini 15–22 Fr; ultra-mini 11–14 Fr; micro <10 Fr; Jackman 1998.
- Fewer transfusions and shorter stay, but longer operation.
- Watch intrarenal pressure; advantage lost against 24 Fr tracts.

IV-3e. Andropause

Type: SAQ (10 marks)

Read as: late-onset hypogonadism (testosterone deficiency of ageing men).

Definition

- **Symptoms plus consistently low serum testosterone** in an ageing man.
- "Andropause" is a misnomer: the fall is gradual and partial. Preferred terms: PADAM (partial androgen deficiency of the ageing male), LOH (late-onset hypogonadism).

Pathophysiology

- Mean testosterone in the seventh decade is about **one-third lower** than in young men.
- **Testis**: fewer Leydig cells, lower sperm output; FSH (follicle-stimulating hormone) rises about threefold.
- **Centre**: blunted feedback, irregular GnRH (gonadotrophin-releasing hormone) pulses.
- **SHBG (sex hormone-binding globulin) rises with age**, so bioavailable (free + albumin-bound) testosterone falls more. Worse with obesity, diabetes, opioids.

Clinical features

- **Sexual (most specific)**: low libido, ED (erectile dysfunction), fewer morning erections.
- Fatigue, low muscle mass, osteoporosis, anaemia, low mood. **ADAM (Androgen Deficiency in the Aging Male)** questionnaire: sensitive, not specific.

Diagnosis

- **Morning (7–11 am) total testosterone, repeated.** AUA (American Urological Association): **<300 ng/dL** (≈10.4 nmol/L); EAU (European Association of Urology): about 12 nmol/L. Calculated bioavailable testosterone when SHBG is abnormal.
- **LH (luteinising hormone)**: high in primary, low or normal in secondary. Low LH → **prolactin**; pituitary MRI if testosterone <150 ng/dL or prolactin high.
- Baseline PSA (prostate-specific antigen), DRE (digital rectal examination), haematocrit.

Treatment

- Lifestyle: weight loss, exercise, treat comorbidities.
- **TRT (testosterone replacement therapy)** for symptoms + confirmed low level: gels, injections (enanthate every 2–3 weeks; undecanoate every 10–14 weeks), oral undecanoate. Stop if no benefit at about 3 months.
- **Contraindications**: prostate or breast cancer; **wish for fertility** (TRT suppresses spermatogenesis; use hCG (human chorionic gonadotrophin) or clomiphene); raised haematocrit; untreated severe sleep apnoea; recent MI (myocardial infarction) or stroke.

Monitoring

- At 3–6 months, then yearly: testosterone, **haematocrit** (hold if >54%), PSA.

Evidence

- **TRAVERSE (2023)**: TRT non-inferior for major cardiac events; more atrial fibrillation and pulmonary embolism.
- **Testosterone Trials**: modest sexual benefit, **no cognitive benefit**.

Diagram to draw: hypothalamic–pituitary–testis axis with ageing defects.

Key points to remember:

- Symptoms + low morning testosterone twice (AUA <300 ng/dL).
- SHBG rises with age; LH separates primary from secondary.
- TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

IV-3f. Nephron Saving Surgery (NSS)

Type: SAQ (10 marks)

Definition

- Complete tumour removal with **maximal preservation of functioning parenchyma**, mainly PN (partial nephrectomy).
- **Rationale**: RN (radical nephrectomy) causes CKD (chronic kidney disease), linked to cardiovascular events and death; PN gives equal cancer control with better function.

Indications

Category	Situations
Imperative	Solitary kidney, bilateral tumours, RN would mean dialysis
Relative	Contralateral kidney threatened (stone, stenosis, diabetes, hypertension); hereditary RCC (renal cell carcinoma)
Elective	**cT1a with normal contralateral kidney: standard of care**

- Increasingly used for cT1b–T2 when feasible. AUA (American Urological Association) 2021: RN only if high complexity, no CKD or proteinuria, and post-RN eGFR (estimated glomerular filtration rate) likely >45.

Pre-operative assessment

- Multiphase CT (computed tomography) or MRI; **R.E.N.A.L. nephrometry score** (radius, exophytic, nearness to collecting system, anterior/posterior, location) predicts complexity.

Technique (open, laparoscopic or robotic)

1. Expose kidney; **intraoperative ultrasound** maps the tumour.
2. Hilar control; clamp artery (± vein), selective, or off-clamp.
3. Excise with a thin rim, or **enucleate** along the pseudocapsule (familial, multifocal tumours).
4. Close collecting system and vessels; renorrhaphy; unclamp; drain.

- **Robotic PN**: shorter warm ischaemia and fewer conversions than laparoscopy.

Ischaemia

- Aim warm ischaemia **<25–30 minutes**; **cold ischaemia (ice slush)** if longer is expected, especially in a solitary kidney.
- Function depends mainly on **parenchyma preserved**. Randomised trials: no functional benefit of off-clamp, hypothermia or mannitol with a normal contralateral kidney.
- **Trifecta**: negative margin + no major complication + warm ischaemia ≤25 minutes.

Complications

- **Urine leak** (stent, drain), **bleeding/pseudoaneurysm** (angioembolisation), AV (arteriovenous) fistula, renal insufficiency, hyperfiltration proteinuria (ACE (angiotensin-converting enzyme) inhibitors).

Outcomes

- Local recurrence 1–2% (elective), 3–5% (imperative). Margin width does not matter if negative.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011)**: RN vs PN for ≤5 cm: PN better function, similar oncology; overall survival slightly

better with RN, but trial flawed.

- **Also know:** thermal ablation for <3 cm; surveillance in frail patients.

Diagram to draw: tumour excised with rim, clamp on renal artery, two-layer renorrhaphy.

> **Key points to remember:**
>
> - Imperative, relative, elective; elective PN is standard for cT1a.
> - Warm ischaemia ≤25 min; preserved parenchyma matters most.
> - EORTC 30904: PN better function, similar cancer control.

Rapid Revision Sheet – Paper IV

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Direct pathway drives early TCMR; indirect pathway drives antibody and chronic rejection.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
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- EORTC 30904: PN better function, similar cancer control.